

# Ella K Say

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## RESEARCH INTERESTS

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I study how neural population dynamics support learning and memory, combining large-scale recordings with computational models to uncover the structure of adaptive behavior.

## EDUCATION

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**University of Washington** Sep 2026 -  
*PhD Neuroscience; NSF Graduate Research Fellow*  
**University of California, San Diego** Sep 2020 - June 2024  
*B.S. Biology with Specialization in Bioinformatics, Minor in Computer Science*

## RESEARCH EXPERIENCE

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**Giocomo Lab, Stanford University** | *Life Science Research Professional I* August 2024 - Present  
• Researching how the hippocampus flexibly encodes features of **spatial memory** and **navigation**.  
• Designed and implemented **virtual reality** paradigms combined with **in vivo 2-photon calcium imaging**.  
**Tye Lab, Salk Institute for Biological Studies** | *Undergraduate Researcher* Mar 2023 - June 2024  
• Investigating the neural mechanisms underlying the **overlap between social and physical pain**.  
• Built **novel machine learning pipelines** to analyze naturalistic social behavioral data.  
**McVicker Lab, Salk Institute for Biological Studies** | *Undergraduate Researcher* Sep 2022 - Mar 2023  
• Developed **ATAC-seq pipelines** to track **chromatin accessibility**.  
• Optimized the **WASP2 pipeline** to facilitate pinpointing regulatory elements that contribute to complex human disease.

## HONORS & ACCOLADES

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**Graduate Research Fellowship** | *National Science Foundation* 2026  
**T32 Training Grant** | *NIH* 2026  
**Travel Award** | *ABRCMS Conference* 2025  
**Oral Presentation Award** | *Louis Stokes CAMP Symposium* 2024  
**Best Poster Award** | *SACNAS National Diversity in STEM Conference* 2023  
**Travel Award** | *SACNAS National Diversity in STEM Conference* 2023  
**CAMP Scholarship** | *National Science Foundation* 2023  
**Provost Honors** | *UC San Diego* 2020-2024

## PUBLICATIONS

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### Manuscripts

1. M. H. Plitt\*, K. Kaganovsky\*, **E. Say**, M. Sosa, T. C. Südhof, L. M. Giocomo, "Hippocampal place code plasticity in CA1 requires postsynaptic membrane fusion," *Neuron*, April 2026.
2. C. Jia\*, A. Tran\*, F. Aloboudi, **E. Say**, N. Thao, C. R. Lee, K. Batra, A. Nguyen, A. Bal, N. N. Nono, J. Delahanty, M. G. Chan, L. R. Keyes, R. R. Patel, R. Wichmann, F. Taschbach, Y. Li, M. Benna, T. D. Pereira, H. Li, K. Tye, "Social Exclusion Amplifies Behavioral Responses to Physical Pain via Insular Neuromodulation," *In revision* 2025.
3. C. Jia\*, A. Tran\*, F. Aloboudi, **E. Say**, N. Thao, C. R. Lee, K. Batra, A. Nguyen, A. Bal, N. N. Nono, J. Delahanty, M. G. Chan, L. R. Keyes, R. R. Patel, R. Wichmann, F. Taschbach, Y. Li, M. Benna, T. D. Pereira, H. Li, K. Tye, "Social Exclusion Amplifies Behavioral Responses to Physical Pain via Insular Neuromodulation," *bioRxiv* May 13, 2025.

### Conferences

1. **E. Say\*\***, C. Dong, L. Giocomo, "Investigating hippocampal coding of spatial and social features through a head-fixed setup," *ABRCMS*, November 2025.
2. **E. Say**, C. Dong, L. Giocomo, "Investigating hippocampal coding of spatial and social features through a head-fixed setup," *Society for Neuroscience*, November 2025.
3. M. H. Plitt\*, K. Kaganovsky\*, **E. Say\*\***, M. Sosa, T. C. Südhof, L. M. Giocomo, "Hippocampal place code plasticity in CA1 requires postsynaptic membrane fusion," *Society for Neuroscience*, November 2025.
4. C. Jia\*, A. Tran\*, F. Aloboudi, **E. Say**, N. Thao, C. R. Lee, K. Batra, A. Nguyen, A. Bal, N. N. Nono, J. Delahanty, M. G. Chan, L. R. Keyes, R. R. Patel, R. Wichmann, F. Taschbach, Y. Li, M. Benna, T. D. Pereira, H. Li, K. Tye, "Social Exclusion Amplifies Behavioral Responses to Physical Pain via Insular Neuromodulation," *Computational and Systems Neuroscience (COSYNE)*, April 2025.
5. C. Jia, A. Tran, F. Aloboudi, **E. Say**, N. Thao, C. R. Lee, K. Batra, A. Nguyen, A. Bal, N. N. Nono, J. Delahanty, M. G. Chan, L. R. Keyes, R. R. Patel, R. Wichmann, F. Taschbach, Y. Li, M. Benna, T. D. Pereira, H. Li, K. Tye, "Social Exclusion Modifies the Neural Response to Physical Pain," *Society for Neuroscience*, November 2023.
6. **E. Say\*\***, C. Jia, A. Tran, N. Thao, K. M. Tye, "Social Exclusion Modifies the Behavioral Response to Physical Pain," *National Diversity in STEM Conference*, November 2023.
7. C. Jia, A. Tran, F. Aloboudi, **E. Say**, N. Thao, C. R. Lee, K. Batra, A. Nguyen, A. Bal, J. Delahanty, M. G. Chan, L. R. Keyes, R. R. Patel, R. Wichmann, F. Taschbach, Y. Li, M. Benna, T. D. Pereira, H. Li, K. Tye, "Social Exclusion Modifies the Neural Response to Physical Pain," *Gordon Research Conference: Modulation of Neural Circuits and Behavior*, May 2023.

\* indicates equal contribution, \*\* indicates presenter

## RESEARCH SKILLS

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### Programming:

- Advanced: Python, MATLAB, Unix/Bash
- Intermediate: C++, Java

**Machine learning:** tensorflow, CUDA, SLEAP, DeepLabCut, keypoint-MoSeq, B-SOID

**Calcium Imaging:** Inscopix scopes, 2-photon imaging, Suite2P, Cellular resolution calcium imaging processing (CNMFE)

**Behavior:** head-fixation, virtual reality paradigm, social behavioral assays, pain behavioral assays, tube testing, IACUC-certified animal handler

**Histology:** Microtome + cryostat slicing, Confocal + episcopes imaging, perfusion and brain extraction

## RELEVANT COURSEWORK

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**Bioinformatics:** Bioinformatics Algorithms, Applied Genomic Technologies, Advanced Bioinformatics Lab, Bioinformatics Statistics, Linear Algebra.

**Computer Science:** Design and Analysis of Algorithms, Advanced Data Structures, Object Oriented Design.

**Neuroscience:** Computational Models in Neuroscience, Structural Biochemistry, Molecular Biology, Neurobiology of Cognition.